Basic properties of relations

For $R \subseteq A^2$, reflexivity means $\forall a \in A, (a, a) \in R$; symmetry means $\forall a, b, aRb \Rightarrow bRa$; and transitivity means $\forall a, b, c, (aRb \land bRc) \Rightarrow aRc$. Irreflexivity forbids every loop and is not the logical negation of reflexivity. Antisymmetry requires $(aRb \land bRa) \Rightarrow a = b$, and differs from asymmetry. For example, $\leq$ is reflexive and transitive without being symmetric.

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Asymmetry differs from antisymmetry

Asymmetry requires $aRb \Rightarrow \neg bRa$, including $a = b$, so it forbids loops. It is equivalent to antisymmetry plus irreflexivity. The first forbids opposite edges between distinct points; the second forbids the same-point case.

Equality $\Delta_A = \{(a, a) : a \in A\}$ is both symmetric and antisymmetric. On a nonempty set, $\leq$ is antisymmetric but not asymmetric. The empty relation on nonempty A is symmetric, antisymmetric, transitive, and irreflexive, but not reflexive.

Transitivity allows repeated variables: aRb and bRa force aRa. Checking only three distinct points misses this. On the empty set the unique relation is both reflexive and irreflexive, since neither universal assertion has a counterexample.

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Counting finite relations

For $|A| = n$, each ordered pair is optional, giving 2^{n^2} relations.

Requirement	Independent choices	Count
Reflexive	Fixed diagonal; other pairs free	$2^{n(n-1)}$
Symmetric	Free diagonal; opposite directions chosen together	$2^{n+\binom{n}{2}}$
Reflexive and symmetric	Fixed diagonal; free unordered pairs	$2^{\binom{n}{2}}$
Antisymmetric	Free diagonal; neither, left, or right per distinct pair	$2^n 3^{\binom{n}{2}}$
Asymmetric	Same choices without diagonal	$3^{\binom{n}{2}}$
Symmetric and antisymmetric	Only diagonal pairs optional	2^n

Adding reflexivity in the last row leaves equality alone. Transitivity couples choices of different pairs and prevents independent multiplication.

Composition direction and matrices

Use the usual function direction. For $R \subseteq A \times B, S \subseteq B \times C$,

$$(a, c) \in S \circ R \iff \exists b\ (aRb \wedge bSc).$$

Apply R first, then S. Some references use left-to-right notation; the pair definition determines the convention.

Both associative expressions mean $\exists b, c\ (aRb \wedge bSc \wedge cTd)$, proving $T \circ (S \circ R) = (T \circ S) \circ R$. Reversing the route gives $(S \circ R)^{-1} = R^{-1} \circ S^{-1}$.

With source rows and destination columns, $M_R \odot M_S = M_{S \circ R}$. Integer multiplication counts two-step walks; Boolean multiplication asks whether at least one exists. For the following properties, take $R \subseteq A \times A$, let $M = M_R$ use the same ordering for rows and columns, and let I be the identity matrix of that order. Entrywise, reflexivity means $I \leq M$, symmetry $M = M^T$, transitivity $M \odot M \leq M$, and antisymmetry $M \wedge M^T \leq I$.

Zero-one matrices

Define $M_{ij} = 1$ iff $(a_i, a_j) \in R$. Reflexivity means an all-one diagonal; symmetry means $M = M^T$. Boolean composition uses

$$(A \odot B)_{ij} = \bigvee_k (A_{ik} \wedge B_{kj}).$$

The reverse relation has the transposed matrix. Transitive closure is the union of Boolean powers. On $n \geq 1$ vertices, positive reachability uses $R \cup \cdots \cup R^n$: distinct endpoints need at most n−1 edges in a simple path, but a nonempty return cycle may need n. Reflexive-transitive closure uses $I \cup \bigcup_{k=1}^{n-1} R^k$. For $n = 0$, the base set is empty, both R and the identity relation are empty, and $R^+ = R^* = \varnothing$; the displayed $n - 1$ power bound is not used. Boolean addition is OR and multiplication AND, unlike integer multiplication.

When $n = 1$, the union from 1 to 0 is empty, so $R^* = I$.

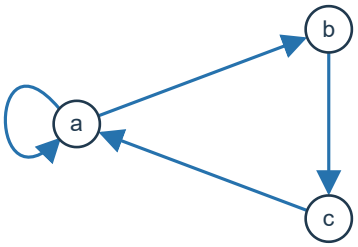
Powers of a three-vertex digraph

Take $a \to a, a \to b, b \to c, c \to a$, indexed a,b,c:

$$M = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \quad M^{\odot 2} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \end{pmatrix}, \quad M^{\odot 3} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}.$$

The square's bb entry is zero because c does not return directly to b. The cube's bb entry is one via $b \to c \to a \to b$; c also first returns in three steps.

Three-point relation: edges and length-2,3 walk endpoints



Original M: 100 OR 010 = row a, 110

M² rows : 111 / 100 / 110

M³ rows : 111 / 110 / 111

b → c → a → b: first positive return

R⁺ = R ∪ R² ∪ R³: all 9 pairs

OR-ing the three powers gives all ones. The square alone is not the closure. Relations are sets of pairs, allowing loops and opposite edges but no parallel copies of a pair.

Closure existence and minimality

Let R^k contain endpoints of exactly length-k walks and $R^0 = \Delta_A$. Define

$$R^+ = \bigcup_{k \geq 1} R^k, \qquad R^* = \Delta_A \cup R^+.$$

Concatenating positive walks proves R^+ transitive and containing R. If transitive T contains R, induction on walk length gives every $R^k \subseteq T$, so $R^+ \subseteq T$. This proves minimality; R* similarly is the smallest reflexive-transitive superrelation.

On $n \geq 1$ vertices, delete internal cycles between distinct endpoints to leave at most n−1 edges. A shortest positive return is a cycle of at most n edges. Hence

$$R^+ = R \cup \cdots \cup R^n, \qquad R^* = \Delta_A \cup \bigcup_{k=1}^{n-1} R^k.$$

The diagonal of a directed three-cycle shows why the first union cannot always stop at n−1.

When $n = 1$, the union from 1 to 0 is empty, so $R^* = I$.

Warshall's invariant and in-place correctness

Start at $W^{(0)} = M_R$ and number vertices 1,…,n:

$$W_{ij}^{(k)} = W_{ij}^{(k-1)} \vee (W_{ik}^{(k-1)} \wedge W_{kj}^{(k-1)}).$$

An entry is one iff a positive i-to-j walk has internal vertices among 1,…,k. A shortest eligible walk either avoids internal k, or splits at k into two segments internally using earlier vertices. Conversely concatenate such segments. If an endpoint is k, a shortest walk needs no internal revisit, so the first term already retains it.

```
W := copy(M)
for k = 1..n:
    for i = 1..n:
        for j = 1..n:
            W[i,j] := W[i,j] OR (W[i,k] AND W[k,j])
```

At fixed k, its row and column update as $x \vee (x \wedge y) = x$. Needed entries therefore stay unchanged, justifying in-place updates. Time is $O(n^3)$, matrix space $O(n^2)$. For R*, initialize the diagonal to one.

Every round on a four-vertex chain

Use edges a→b,b→c,c→d and indices a,b,c,d.

Round	Intermediate	New pairs	Four matrix rows
0	None	Original	0100 / 0010 / 0001 / 0000
1	a	None	0100 / 0010 / 0001 / 0000
2	b	(a,c)	0110 / 0010 / 0001 / 0000
3	c	(a,d),(b,d)	0111 / 0011 / 0001 / 0000
4	d	None	0111 / 0011 / 0001 / 0000

Round three uses the preceding round's a→c to obtain a→d, rather than only original edges. The acyclic graph's R^+ diagonal is zero; R^* adds loops.

Reflexive closure is $R \cup \Delta$; symmetric closure is $R \cup R^{-1}$. Equivalence closure is $(R \cup R^{-1})^*$. Reversing walks gives symmetry. Every equivalence superrelation contains inverses, identities, and concatenations, so contains this expression. Equivalently, ignore directions, find components, and include every within-component pair.

Every round on a four-vertex chain (continued)

The intersection of a **nonempty** family of relations on a fixed set A preserves reflexivity, symmetry, antisymmetry, and transitivity separately. For antisymmetry, if both aRb and bRa lie in the intersection, choose one relation in the family and use its antisymmetry to conclude a=b. Nonemptiness is needed to choose that relation. The other properties follow by applying their conditions in each member.

The empty-family intersection relative to A^2 is A^2. It is reflexive, symmetric, and transitive, but if A has distinct elements a,b it contains both (a,b) and (b,a), so is not antisymmetric. The nonempty-family condition cannot be omitted. Unions need not preserve transitivity: {(1,2)} and {(2,3)} are individually transitive but their union lacks (1,3).

A union need not preserve transitivity

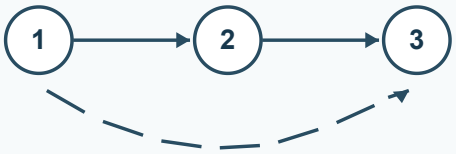
Each is transitive

R = {(1,2)}

S = {(2,3)}

One missing pair disproves transitivity

Union need not be transitive



Missing 1→3

On $A = \{1, 2, 3\}$, let $R = \{(1, 2)\}$ and $S = \{(2, 3)\}$. Neither relation has two composable edges, so each is vacuously transitive. Their union has $1 \to 2 \to 3$ but lacks $(1, 3)$, and is not transitive.

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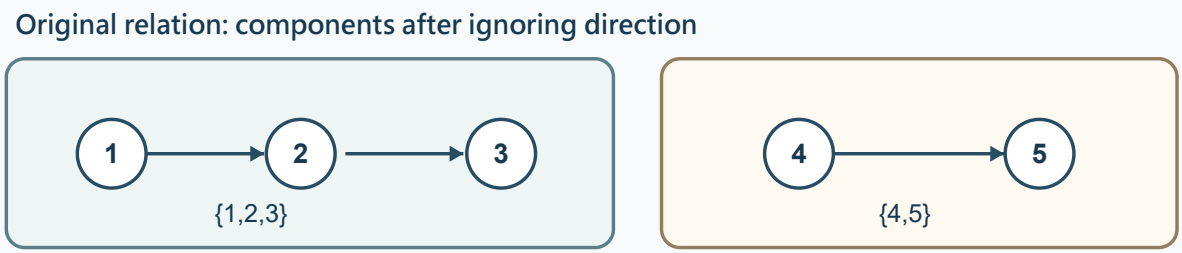
Intersections and the empty family

Suppose each $R_i \subseteq A^2$ is transitive. If $(a, b), (b, c) \in \bigcap_{i \in I} R_i$, both pairs occur in every R_i. Transitivity puts (a, c) in every R_i, hence in the intersection. This proves the claim for a nonempty family.

For $I = \varnothing$, define the intersection relative to A^2 as A^2; it is still transitive. However, if $|A| \geq 2$, it is not antisymmetric. Thus transitivity is preserved even for the empty family, while antisymmetry requires this boundary to be considered separately.

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Equivalence closure via connected components



Let $A = \{1, 2, 3, 4, 5\}$ and $R = \{(1, 2), (2, 3), (4, 5)\}$. Ignoring direction gives components $B_1 = \{1, 2, 3\}$ and $B_2 = \{4, 5\}$. The candidate closure is $E = (B_1 \times B_1) \cup (B_2 \times B_2)$.

Why is the closure minimal?

The candidate E relates points in the same component. It is reflexive, symmetric and transitive, and it contains every edge of R.

Conversely, any equivalence relation containing R must contain reversed edges and self-pairs, and transitivity forces it to contain both endpoints of every undirected path. It therefore contains all pairs within both components, hence contains E. Thus E is the smallest equivalence relation containing R, with $3^2 + 2^2 = 13$ ordered pairs and no pairs across components.

Explicitly, $B_1^2 = \{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3)\}$ and $B_2^2 = \{(4, 4), (4, 5), (5, 4), (5, 5)\}$.

Partial orders

A partial order is reflexive, antisymmetric, and transitive; $(A, \preceq)$ is a poset. Two elements are comparable if $a \preceq b$ or $b \preceq a$, otherwise incomparable. A total order makes every pair comparable.

Write $a \prec b$ for $a \preceq b, a \neq b$. Element b covers a if $a \prec b$ and no c satisfies $a \prec c \prec b$. The word partial means not every pair must be comparable.

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Strict orders, restrictions, and covers

If $a \prec b \prec c$, transitivity gives $a \preceq c$. Were a=c, antisymmetry would force a=b, a contradiction. Thus the strict relation is irreflexive and transitive. Conversely, adding equality to an irreflexive transitive relation gives a partial order, since opposite comparisons would force a loop.

Restriction to $B \subseteq A$ preserves the three axioms but changes covers. Removing 2 from 1≺2≺4 makes 1–4 a cover. In a finite poset, insert intermediate elements in any noncover a≺b; finiteness prevents endless insertion, giving a chain of covers. The infinite dense order $(\mathbb{Q}, \leq)$ has no covers, so an empty cover graph cannot reconstruct it.

Positive-integer divisibility is a partial order. On nonzero integers, 1 and −1 divide each other without equality, giving only a preorder.

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Hasse diagrams

Place smaller elements lower, draw all covers, omit loops, and represent transitivity by upward paths. Only covers receive direct edges; the upward convention permits omitting arrows. If a⩽b has an intermediate c, a–b is not a Hasse edge. Drawing every pair in transitive closure adds redundant lines and hides cover structure.

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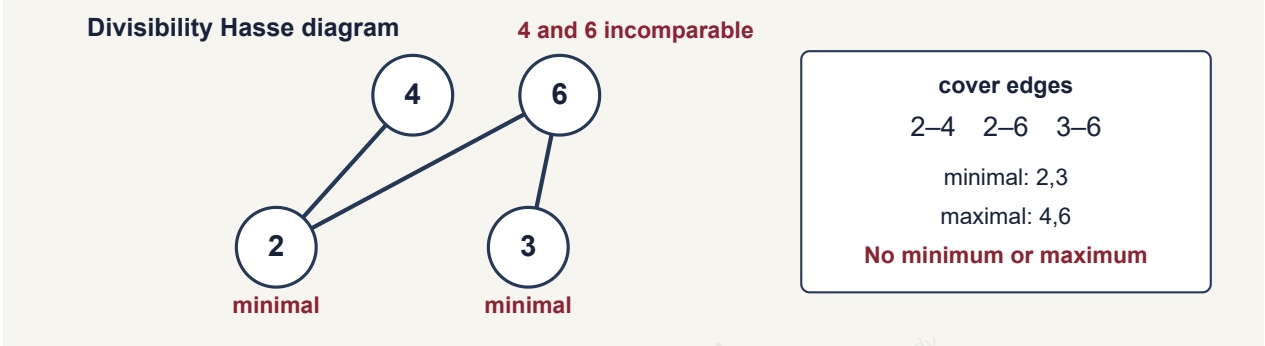
Extremal elements

A minimal element $m \in S$ has no $x \in S$ with $x \prec m$; a maximal element has no strictly larger element. A minimum must satisfy $m \preceq s$ for every $s \in S$, and a maximum requires $s \preceq M$ for every $s \in S$.

There may be several minimal or maximal elements. A minimum or maximum, when it exists, is unique. A minimum is minimal, but minimality alone does not require comparability with every element.

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Extremal elements in a divisibility order



For $A = \{2, 3, 4, 6\}$, the three edges are exactly the covers. The minimal elements are 2 and 3; the maximal elements are 4 and 6. The two minimal elements and the two maximal elements are respectively incomparable, so there is neither a minimum nor a maximum.

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Finite extrema and a unique maximal element

Every finite nonempty poset has a maximal element. Start anywhere and move strictly upward whenever possible. No vertex repeats, since that would imply x≺x; after at most |A| −1 moves the process stops. Descending similarly finds a minimal element.

Greatest elements are unique: candidates u,v satisfy u⩽v and v⩽u, hence u=v. The least case is dual. If a finite poset has a unique maximal m, ascent from any x ends at a maximal, necessarily m. Thus x⩽m for every x and m is greatest.

Finiteness matters. On $\mathbb{N} \sqcup \{z\}$, give naturals their usual order and compare z only with itself. Naturals always have larger elements; z is uniquely maximal but not greatest. The empty poset has no extrema.

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Bounds may lie outside the subset

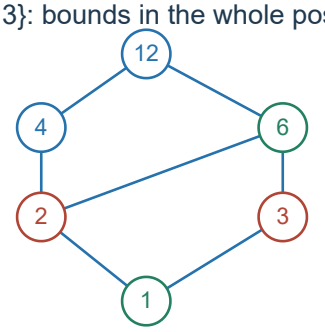
In a poset A, a lower bound l of $S \subseteq A$ satisfies $l \preceq s$ for every $s \in S$; an upper bound u satisfies $s \preceq u$. Bounds are sought in all of A, not necessarily in S.

The greatest lower bound is the maximum among lower bounds; the least upper bound is the minimum among upper bounds. Antisymmetry makes each unique when it exists.

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Common bounds among the divisors of 12

S={2,3}: bounds in the whole poset; glb=1, lub=6



Upward means divides; only covers are drawn.

S: red 2,3 (incomparable)

Common lower bounds: {1}; glb=1

Common upper bounds: {6,12}; lub=6

6 ≤ 12; not every upper bound is the lub.

For $S = \{2, 3\}$, the lower bounds are $\{1\}$ and upper bounds are $\{6, 12\}$. Thus glb = 1 and lub = 6, both outside S; S itself has neither a minimum nor a maximum.

In the divisibility order of all positive integers, meet is gcd and join is lcm. Deletion can destroy bounds: in $\{2, 3, 6, 12\}$, 2 and 3 have no common lower bound.

A lattice's universal properties

A nonempty poset is a lattice when every pair has meet a∧b and join a∨b. Common bounds are independent of the pair's order, proving commutativity.

Common lower bounds of a,a∨b are exactly lower bounds of a: one direction requires being below a; the other uses x⩽a⩽a∨b. Hence $a \wedge (a \vee b) = a$. Reversing the order gives dual absorption. Both bracketings of a three-element meet are the glb of {a,b,c}; uniqueness proves associativity, and likewise for joins.

Repeated meet/join over a finite lattice yields bottom/top. Order reversal interchanges upper/lower, minimal/maximal, join/meet. Top and bottom alone need not give a lattice; a lattice need not be distributive. Chapter 15 adds distribution and complements, so Boolean conclusions do not apply to arbitrary posets.

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Topological sorting with proof and trace

A finite strict order is a DAG: a cycle would imply $a \prec a$. Conversely, reflexive reachability in a DAG is a partial order because distinct mutually reachable vertices would form a cycle.

Repeatedly output and remove a currently minimal element. Finite descent guarantees one exists. If originally $a \prec b$, then b cannot be minimal until a is removed, so the resulting linear extension preserves every comparison.

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Topological sorting the divisors of 12

At each step choose the numerically smallest current minimum:

Already output	Current minima	Choice
None	{1}	1
1	{2,3}	2
1,2	{3,4}	3
1,2,3	{4,6}	4
1,2,3,4	{6}	6
1,2,3,4,6	{12}	12

Cover adjacency lists and indegrees allow initialization and processing each edge once in $O(n+m)$ time. This excludes finding covers from a full relation matrix.

Linear extensions and Catalan words

For an integer $n \geq 1$ and distinct primes p,q, divisors of $p^{n-1}q$ form chains $A_i = p^i$, $B_i = p^i q$, 0≤i<n. Each chain increases with i; across chains, $A_i \preceq B_j$ iff $i \leq j$, and no B lies below an A.

Record an extension as n A's and n B's. Within each chain, occurrence order determines indices uniquely. Legality is equivalent to every prefix having at least as many A's as B's. Before the (j+1)st B, A_j must appear; that also ensures all A_i for $i \leq j$ have appeared. The reflection bijection of Chapter 1 gives

$$C_n = \binom{2n}{n} - \binom{2n}{n-1} = \frac{1}{n+1}\binom{2n}{n}.$$

For n=3, the five words are AAABBB,AABABB,AABBAB,ABAABB,ABABAB. Replace successive A's by 1,2,4 and B's by 3,6,12 to recover all five extensions of the divisor poset of 12.

Equivalence relations and classes

A reflexive, symmetric, transitive relation is an equivalence relation. Define $[a] = \{x : x \sim a\}$. Classes are equal or disjoint and partition A.

If x∈[a]∩[b], symmetry/transitivity give a~b. Any y∈[a] then satisfies y~b, proving [a]⊆[b]; reversing roles proves equality. A representative may label a class arbitrarily, but the class itself is a fixed set.

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Integers modulo five

Define a~b iff a−b=5k for an integer k. Reflexivity uses k=0; symmetry negates k; transitivity adds two multiples of five. The classes are

$$[r] = \{5k + r : k \in \mathbb{Z}\}, \qquad r = 0, 1, 2, 3, 4.$$

They partition the integers. The quotient is $\mathbb{Z}/5\mathbb{Z}$. Chapter 14 adds ring operations, whose well-definedness must be proved before asserting that additional algebraic structure.

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Equivalence and partition are inverse constructions

Classes cover A since a∈[a]. If x∈[a]∩[b], a~x~b gives a~b. Every y~a then satisfies y~b, proving inclusion and, by reversal, equality. Thus a~b iff [a]=[b], and distinct classes are disjoint.

From a partition, relate elements in the same block. Each element's unique block proves reflexivity; symmetry is immediate. If a,b∈B and b,c∈C, the common b forces B=C, proving transitivity. The block containing a is precisely [a].

Taking classes and then the same-block relation recovers the original relation. Taking the relation and then classes recovers the original blocks. These are inverse bijections. The empty partition has no blocks, including no empty block, and corresponds to the empty relation on the empty set.

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Representative independence on quotients

For f:A→B define $a \sim_f b$ iff f(a)=f(b). Equality's properties give equivalence; projection π(a)=[a] is onto the quotient. Set $\bar{f} : A/\!\sim_f \to f(A)$, $\bar{f}([a]) = f(a)$. Changing representatives leaves the value unchanged, so it is well-defined. Equal values imply $a \sim_f b$ and equal classes, proving injection; every image has a preimage, proving surjection. Therefore

$$A/\!\sim_f \longleftrightarrow f(A), \qquad f = \bar{f} \circ \pi.$$

Generally g factors as $\bar{g} \circ \pi$ iff it is constant on every class. Equal projection values prove necessity; defining $\bar{g}([a]) = g(a)$ proves sufficiency. Surjectivity of π makes the factor unique.

Modulo five, [a]↦a mod 5 is well-defined; [a]↦a^2∈Z is not, since [1]=[6] but 1≠36. Quotient operations require the same representative check.

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Class sizes, counts, and refinement

For class sizes $s_1, \ldots, s_k$, $R = \bigcup_i (B_i \times B_i)$ is a disjoint union. Thus $|R| = \sum_i s_i^2$, with $\sum_i s_i = n$, giving $n \le |R| \le n^2$ and $|R| \equiv n \pmod 2$. Range and parity are only necessary conditions.

S(n,k) counts partitions into k unlabeled nonempty blocks. A new element either forms a singleton or joins one of k old blocks, giving

$$S(n,k) = S(n-1,k-1) + kS(n-1,k), \qquad S(0,0) = 1.$$

Out-of-range values are zero. The Bell number $\sum_k S(n,k)$ counts equivalence relations.

Write $P \preccurlyeq Q$ when every P-block lies in a Q-block. Inclusion gives reflexivity/transitivity. Under mutual refinement, $B \in P$ lies in $C \in Q$, itself in $D \in P$. Nonempty disjoint blocks force $B = D$, then $B = C$; hence $P = Q$. This order is equivalent to $R_P \subseteq R_Q$.

DFA length invariant and recurrence

For a complete DFA $(Q, \Sigma, \delta, q_0, F)$, define

$$p \equiv_k q \iff \forall |w| \le k : [\delta^*(p,w) \in F \iff \delta^*(q,w) \in F].$$

Empty input gives $P_0 = \{F, Q \setminus F\}$, omitting empty blocks. A nonempty word is av, with $|v| \le k$, so

$$p \equiv_{k+1} q \iff (p \equiv_0 q) \wedge \bigwedge_{a \in \Sigma} [\delta(p,a) \equiv_k \delta(q,a)].$$

Partitions only get finer. Signatures may include the current block label and each input's successor-block label; retaining the current label does not over-refine because the new relation is contained in the old.

At a fixed point $P_{k+1} = P_k$, same-block states have same-block successors under every input. Induction on suffix length proves all-word agreement: empty input agrees, and av reaches equivalent successors before applying the induction hypothesis to v. Conversely behaviorally equivalent states never split at any finite round. The stable partition is exactly behavioral equivalence.

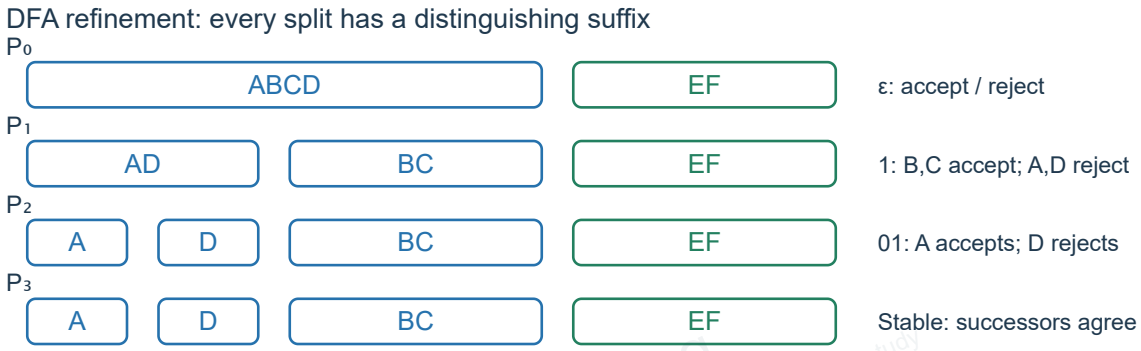
A complete six-state refinement trace

Start at A, accept $\{E, F\}$, and use alphabet $\{0, 1\}$:

State	0	1	Accepts empty input?
A	B	C	No
B	D	E	No
C	D	F	No
D	D	D	No
E	E	E	Yes
F	E	F	Yes

The words $\varepsilon, 0, 1, 00, 01, 11$ reach A, B, C, D, E, F respectively, so all six states are reachable.

Each DFA refinement round



P_1 splits ABCD into AD and BC according to whether input 1 leads to an accepting state. P_2 splits A and D because input 0 reaches different P_1 blocks; $P_3 = P_2$. Stability requires every state in each block to transition into one block for every input.

Quotient DFA and the lower bound

Stable blocks U={A}, V={B,C}, D={D}, T={E,F} give representative-independent $\delta'([p], a) = [\delta(p, a)]$.

Block	0	1	Accepting?
U	V	V	No
V	D	T	No
D	D	D	No
T	T	T	Yes

Empty suffix distinguishes T from each other block; 1 distinguishes U,V and V,D; 01 distinguishes U,D. If v distinguishes successors under a, av distinguishes original states. First-split rounds therefore recover witnesses backward.

Choose an access prefix for each reachable class. If another same-language DFA sends prefixes x,y for two distinct classes to one state, every common suffix has the same acceptance there. A witness w makes exactly one of xw,yw accepted in the original, contradiction. Every equivalent DFA needs at least this many states; the quotient attains the bound.

Refinement cost and unreachable states

Each genuine refinement adds a block, so at most $n - |P_0|$ occur. Signatures cost $O(n|\Sigma|)$ per round, giving $O(n^2|\Sigma|)$ with array labels and linear grouping; comparison sorting adds its own cost.

For minimization from a specified initial state, first remove unreachable states by graph search. Otherwise classes describe every state's behavior, including classes the initial state never reaches. For a partial DFA, add one common rejecting sink and complete all missing transitions first. Missing edges cannot alternate between rejection and ignoring input.

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Mealy machines have no accepting partition

A Mealy machine $(Q, \Sigma, \Gamma, \delta, \lambda)$ emits λ(q,a) on input a, then moves to δ(q,a). Its entire output satisfies

$$\lambda^*(q, \epsilon) = \epsilon, \quad \lambda^*(q, av) = \lambda(q, a)\lambda^*(\delta(q, a), v).$$

Equivalence compares whole outputs for every word and has no accepting set. Let $p \approx_k q$ mean agreement for lengths ≤k. P_0 is one block Q, and

$$p \approx_{k+1} q \iff \forall a : [\lambda(p, a) = \lambda(q, a) \wedge \delta(p, a) \approx_k \delta(q, a)].$$

Compare outputs for every input, then successor blocks. DFA P_0 compares empty-word acceptance; Mealy P_1 first compares one-symbol outputs. Do not mix these indices.

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Mealy merging and output witnesses

Each cell gives next state/output; the initial state is p:

State	0	1
p	q/0	r/0
q	q/0	s/1
r	q/0	s/1
s	s/1	s/1

$P_0 = \{pqrs\}$. Output vectors (0,0), (0,1), (1,1) give $P_1 = \{p\}/\{q, r\}/\{s\}$. The respective successor signatures are (qr,qr), (qr,s), (s,s), so $P_2 = P_1$ and q,r merge.

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Distinguishing words and Mealy minimality

Input 1 distinguishes p,q with outputs 0,1; input 0 distinguishes q,s with outputs 0,1. On input 01, p emits 01 and s emits 11.

If t has p's output vector (0,0) but moves to s on input 0, then 00 distinguishes p,t with outputs 00,01. Induction on word length using the output recurrence proves stability correctness.

For minimality, use reachable access prefixes and a distinguishing suffix: different future-output behaviors cannot share a state in an equivalent machine. Compare future outputs after the prefix without mixing in already emitted output.

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A seven-state Mealy machine: outputs before successors

Each cell is “next state / output”. Before choosing an initial state, all seven states belong to $P_0 = \{Q\}$ because the empty input produces the empty word.

State	Input 0	Input 1	One-step output vector (0,1)
1	5 / 1	7 / 0	(1,0)
2	7 / 1	2 / 0	(1,0)
3	6 / 1	1 / 0	(1,0)
4	3 / 0	4 / 0	(0,0)
5	3 / 0	5 / 0	(0,0)
6	2 / 1	7 / 0	(1,0)
7	4 / 1	1 / 0	(1,0)

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Refinement by input length

P1 · Outputs for words of length ≤1
{1,2,3,6,7} | {4,5}

P2 · Outputs for words of length ≤2
{1,7} | {2,3,6} | {4,5}

P3 · Outputs for words of length ≤3
{1,7} | {2} | {3,6} | {4,5}

P4 · Outputs for words of length ≤4
{1,7} | {2} | {3} | {6} | {4,5}

States in the same block of P_k give identical output words for every input of length at most k. At $P_4 = P_5$, successors remain in the same blocks. Induction on input length then gives equal output for every finite input.

How successor signatures split blocks

Within each old block compare $(\lambda(q,0), [\delta(q,0)]), (\lambda(q,1), [\delta(q,1)])$. The table omits the identical one-step outputs within each compared block and lists **old successor blocks** in input order 0,1:

Round	Old blocks	States	Successor blocks (0,1)
$P_1 \to P_2$	X={1,2,3,6,7}, Y={4,5}	1,7; 2,3,6	(Y,X); (X,X)
$P_2 \to P_3$	A={1,7}, B={2,3,6}, C={4,5}	2; 3,6	(A,B); (B,A)
$P_3 \to P_4$	A={1,7}, B={2}, C={3,6}, D={4,5}	3; 6	(C,A); (B,A)

Compare signatures **within the same old block**; blocks already separated by output never merge. States within each other block have equal signatures and do not split.

A shortest distinguishing word and the quotient

From states 3 and 6, read the same input 0000:

Initial state	Four-step state path	Output
3	3→6→2→7→4	1111
6	6→2→7→4→3	1110

The states share a block in P_3, so no input of length at most 3 distinguishes them. The word 0000 has length 4 and yields different outputs; it is **a shortest distinguishing word**, not necessarily the only one.

The stable blocks are A={1,7}, B={2}, C={3}, D={6}, E={4,5}. To minimize the machine for a **specified initial state**, remove states unreachable from it and then quotient the reachable machine. These five blocks classify all original states; without an initial state they do not establish that all five are reachable in the minimized machine.

Quotient transitions and outputs

Use the stable blocks A={1,7}, B={2}, C={3}, D={6}, E={4,5} as states. Each cell is “next block / output”:

Block	Input 0	Input 1
A	E / 1	A / 0
B	A / 1	B / 0
C	D / 1	A / 0
D	B / 1	A / 0
E	C / 0	E / 0

For example, on input 0, representatives 1 and 7 of A go to 5 and 4, both in E, and both output 1; on input 1 they both go to A and output 0. The same check works for each block and input: stability makes the quotient independent of the representative.

When can Mealy states merge?

Check equal outputs for every input and successor membership in the same stable blocks. Induction on input length then gives equal future output streams. A split can be traced through successors to a distinguishing word. Once an initial state is specified, remove unreachable states before taking the quotient.

References and attribution

Lin Cing, handwritten discrete mathematics notes. Ralph P. Grimaldi, Discrete and Combinatorial Mathematics, 5th ed., Ch. 7.

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